

Support-Sensitive Certificates for Unknot Recognition

Optimal coordinate transport, component profiles,
and linear unit-pivot ray certificates

Research continuation prepared for Vladimir Reshetnikov
ProveIt project; mathematical development and implementation with ChatGPT

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Abstract

We develop exact, independently checkable improvements to the normal-surface analysis stage of ProveIt’s unknot recognizer. For a supplied admissible normal vector x , let d be the nullity of the matching equations after all zero coordinates of x have been fixed. We compile a decoder from exactly d selected coordinates, minimize its allocated binary-slot cost by a weighted matroid basis calculation, and transport the selected data as one scalar. A fibre invariant proves that scalar packing preserves every canonical weight run of the corresponding vector calculation. We sharpen the bound on distinct full component-coordinate types to $H(x) \leq 2d - a$, where a counts supported vertex links, and to $H(x) \leq d$ when all components are two-sided.

A second path partitions the source by actual face-arc incidences and counts essential discs on rank-one blocks without an interval-orbit computation. An integral unit-pivot peeling certificate proves rank one and recovers the primitive generator from a single coordinate. For the entire layered Fibonacci meridian family it has exactly $3t - 1$ steps in t tetrahedra, despite exponentially many represented normal pieces. The deliverable contains additive Python implementations, independently written checkers, adversarial tests, a reproducible corpus audit, paired benchmarks, and a diagram-bound positive-certificate interface.

These are local supplied-surface results. Compressed component extraction and primitive vertex-surface connectedness are classical antecedents, explicitly credited here. We do not prove a quasi-polynomial bound for unrestricted unknot recognition. We identify the remaining selection, hierarchy, marking, and representation bounds needed for such a theorem and formulate concrete next research targets.

Reproducibility baseline. [ProveIt](#), commit 3a90fb34146c915328ab8eac6250cc2514f74ed0. All performance statements refer to the accompanying executable experiment, not to an unspecified later revision.

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1 The problem addressed and the contribution boundary

The long-term goal is a deterministic unknot recognizer whose running time, on a diagram with n crossings, is $2^{(\log n)^{O(1)}}$. A useful continuation must improve a concrete part of the algorithm while keeping the distinction between a local polynomial routine and the global search explicit. A normal surface can contain exponentially many elementary discs and yet have a short binary coordinate vector. Operating on those coordinates is indispensable, but does not by itself determine which surface recognition should examine.

This article concentrates on two questions that the maintained implementation already exposes as precise computational interfaces:

1. Given a normal vector, what are the distinct complete coordinate vectors of its connected components, with binary multiplicities?
2. How many of its components are discs with essential boundary on the boundary torus?

The first question needs more information than the second. Our implementation uses a compiled coordinate observer for the first, and a rank-one algebraic shortcut with an exact fallback for the second. An additional wrapper binds a positive answer to the canonical exterior of a particular classical knot diagram. It accepts supplied normal coordinates; it does not search for them.

1.1 What the pinned repository and incoming work already contain

The audit covered the maintained Python implementation and synthesis sources, and the relevant incoming archives at the pinned commit. The maintained coordinate census already transports positive quadrilateral coordinates and one positive minimum triangle anchor per supported vertex link. Its dimension is essentially $q + a$, where q counts positive quadrilateral types. Its decoder is sparse and integral. Replacing it with generic linear algebra is therefore a tradeoff, not an automatic improvement.

The maintained code also contains exact three-weight essential-disc detection, an independent orbit replay checker, primitive-core disc counting, a canonical diagram-to-exterior construction and independent local verification, and minimum-span cocycle optimization with a forced-difference quotient. Earlier synthesis establishes component-type bounds using quadrilateral coordinates. The incoming support-kernel package already develops small supported systems, primitive-ray certificates, low-nullity ray enumeration, and dense Fibonacci examples. We do not claim these as new discoveries. The accompanying provenance manifest names the inspected files and incoming archive hashes.

The additions here are a cost-optimal observer on the *actual positive standard-coordinate support*, a scalar realization with an exact live-state invariant, the $2d - a$ inventory refinement, safe geometric support blocks, and an executable unit-pivot specialization that proves and exploits the integral ray structure without Gaussian elimination. Their composition with existing source correspondence is also implemented and tested.

1.2 Classical and current external context

Agol–Hass–Thurston already give compressed component coordinates for a supplied normal surface in time polynomial in triangulation size and logarithmic normal weight [2, Theorem 16, Corollary 17]. Our observer and packing refine that local calculation. Hass–Lagarias–Pippenger use active-constraint rank and primitivity to certify vertex-surface connectedness without tracing all pieces [1, Lemma 5.3, Sections 6 and 8]. Our ray path specializes that established principle. Weighted-basis optimization is the matroid greedy theorem [3]; the contribution is the exact objective and certified application to this implementation.

Lackenby’s inspected 2021 Oxford handout announces $2^{O((\log n)^3)} = n^{O((\log n)^2)}$ [4, pp. 7, 25]. The July 2026 preprint gives new hierarchy and certification results. Its step-visit bound is $L(g+1)^L$, but individual step costs are not all specified; it does not assert a quasi-polynomial total bound [5, Sections 9–10]. We use this distinction to formulate a precise global accounting target in section 11, rather than infer a complexity theorem from a local benchmark.

2 Input model, support, and what a certificate means

Let $\mathcal{T}$ be a finite triangulation with t tetrahedra, given by face pairings. For disc queries we require the maintained validator’s contract: a connected compact orientable 3-manifold, one torus boundary component, and spherical or disc vertex links. The linear and component-profile arguments need the usual normal-coordinate uniqueness and connected vertex links; they do not need a knot-diagram origin. The source-bound interface adds that origin separately.

Let $x \in \mathbb{Z}_{\geq 0}^{7t}$ satisfy the matching equations and quadrilateral constraints. Write $F(x)$ for its normal realization, unique up to normal isotopy [1, Theorem 5.1]. Each connected component has a nonzero vector y with $0 \leq y \leq x$. Set

$$S = \{i : x_i > 0\}, \quad s = |S|, \quad A = A_S, \quad r = \text{rank}_{\mathbb{Q}} A, \quad d = s - r, \quad \mathcal{K}_S = \ker_{\mathbb{Q}} A. \quad (1)$$

Here A_S restricts the full matching matrix to the columns in S . There are at most $6t$ rows and at most $5t$ supported columns. Every original row has the form $T + Q - T' - Q' = 0$. After coincident terms cancel, its nonzero coefficients are ± 1 and its Euclidean norm is at most 2.

Since x is positive on S , the nonnegative cone in $\mathcal{K}_S \otimes \mathbb{R}$ has relative dimension d . Indeed a small neighbourhood of x within this space stays positive. Thus d is the actual dimension of the supported solution face, not merely a dimension of a larger quadrilateral sector. For $x = 0$ we use $S = \emptyset$ and $s = r = d = 0$.

Let $B = \max_{i \in S} \text{bits}(x_i)$, with $B = 0$ for the empty source, and let $W = \sum_i x_i$. Binary input length is bounded above by $O(t \log(t+1) + sB)$ bits, with actual coordinate length $\sum_{i \in S} \text{bits}(x_i)$; $\log(W+1) = O(B + \log(t+1))$. Unit-cost arithmetic on a B -bit integer is not a bit-complexity model. We state structural operation counts separately from costs of reading, adding, multiplying, or decoding these integers.

Definition 2.1 (Component profile). The full component profile is the finite map sending each distinct complete $7t$ -coordinate vector y of a connected component to its multiplicity in $F(x)$. Its support size is $H(x)$. Equal homeomorphism types with different normal coordinates are distinct entries.

A certificate in this work is an exact finite object checked against the original triangulation and source coordinates. Producers can be complicated or heuristic. Acceptance depends on a separately implemented replay of algebraic identities and the maintained finite geometry and orbit contracts. This is an independently checked implementation, not a proof-assistant formalization or a claim that the entire Python runtime is formally verified.

3 An optimal coordinate observer on the source support

3.1 Dimension and minimum binary-slot cost

For $i \in S$ put $b_i = \text{bits}(x_i) = \lceil \log_2(x_i + 1) \rceil$. A retained coordinate needs a slot of this width to hold every value in $[0, x_i] \cap \mathbb{Z}$. For $J \subseteq S$, let π_J be coordinate restriction.

Theorem 3.1 (Minimum-cost support projection). *The map π_J is injective on $\mathcal{K}_S$ if and only if the columns of A indexed by $S \setminus J$ are independent. In particular, an injective coordinate*

projection of minimum total cost $\sum_{j \in J} b_j$ retains exactly d coordinates. A minimum is obtained by greedily taking independent pivot columns in nonincreasing order of b_i and retaining their complement.

Proof. A nonzero vector in $\ker A$ with zero restriction to J is exactly a nonzero linear dependence among the complementary columns. This proves the first claim. All costs are positive, so a cost-minimizing complement extends to a column basis; otherwise removing another retained coordinate decreases cost. Its size is r , and the retained size is d .

Minimizing the retained sum is equivalent to maximizing the pivot sum, because their sum is the fixed total $\sum_{i \in S} b_i$. Column independence forms a representable matroid. For completeness, take a maximum-weight basis containing the current greedy prefix. If the next accepted greedy column is absent, the fundamental circuit obtained by adding it contains a basis column outside the prefix that may be exchanged out. Such a column has no greater weight: an earlier rejected column was already spanned by an earlier prefix. The exchange preserves maximum weight and extends the greedy prefix. Induct until the greedy basis is installed. Its complement gives the desired minimum. Ties are broken by coordinate index for reproducibility. $\square$

Corollary 3.2 (Comparison with the maintained observer). *Let q count positive quadrilateral types and let a count supported vertex links. Then $d \leq q + a$, and the new minimum slot budget is no greater than that of the maintained positive-quadrilateral/vertex-anchor observer.*

Proof. Retain all positive quadrilaterals and one triangle coordinate for each fully supported vertex link. A kernel vector vanishing on these coordinates has zero quadrilaterals. Triangle matching then makes its triangle heights constant around each connected vertex link. That constant is zero either because an unsupported triangle was fixed to zero or because its retained anchor is zero. Thus the retained projection is injective. Apply theorem 3.1 to this feasible coordinate set. $\square$

The optimality domain matters. This minimizes allocated binary slots among coordinate projections admitting a linear decoder on the whole supported matching space. It does not optimize arbitrary finite encodings, decoder sparsity, or total runtime. A single huge integer can encode a finite box. Nor is d necessarily the dimension of the finite set of components appearing in this one source. For genuine mixed-radix bases $x_j + 1$, minimizing their product instead uses greedy weights $\log(x_i + 1)$, equivalently the order of x_i . The shipped implementation optimizes binary slots.

3.2 Exact decoder and independent rank evidence

Let $P = S \setminus J$ be the pivot basis. Choose r original rows R for which $M = A[R, P]$ is nonsingular, put $C = A[R, J]$, and let $\Delta = \det M \neq 0$. The rational decoder is

$$T[J, :] = I_d, \quad T[P, :] = -M^{-1}C, \quad D = |\Delta|, \quad N = DT. \quad (2)$$

It sends retained values w to Nw/D . We need not ask a checker to trust the elimination that produced N .

Theorem 3.3 (Certified support decoder). *Suppose a certificate supplies complementary sets J, P , rows R , an integer $D > 0$, an integer matrix N , and evidence that $A[R, P]$ is nonsingular. If*

$$|J| = d, \quad |P| = |R| = s - d, \quad N[J, :] = DI_d, \quad AN = 0,$$

then $\dim \ker A = d$ and every $y \in \ker A$ equals Ny_J/D . For standard normal matching matrices a certificate of this form exists with

$$1 \leq D \leq 2^r, \quad |N_{ij}| \leq 2^r.$$

Proof. The identity rows make the d columns of N independent; $AN = 0$ puts them in the kernel. Hence nullity is at least d . The nonsingular $(s - d)$ -minor gives rank at least $s - d$, so equality holds in both bounds. If y is in the kernel, $y - Ny_J/D$ is also in the kernel and vanishes on J . The pivot minor forces its remaining coordinates to vanish.

For existence use eq. (2). Cramer's rule expresses each entry of N as a signed r -minor, with at most one pivot column replaced by a retained column. Every row of each such minor has norm at most 2. Hadamard's inequality bounds its determinant by 2^r . The identity rows obey the same bound. When $r = 0$, take $D = 1$, $N = I_s$, and the empty minor. $\square$

The rank evidence consists of a modulus $m \geq 2$ and the claimed minor. The checker performs its own modular elimination and only inverts pivots whose greatest common divisor with m is 1. Successful elimination makes the determinant a unit modulo m , and thus a nonzero integer. Primality is not a soundness assumption. In particular, merely testing that a pivot is nonzero modulo a composite modulus would be insufficient.

The producer chooses a small prime not dividing D . Here is a self-contained polynomial value bound. For $r \geq 1$ put $M_0 = 16(r + 1)^2$. If every prime at most M_0 divided $D \leq 2^r$, there would be at most r such primes, and $\text{lcm}(1, \dots, M_0) \leq M_0^r$. The central binomial coefficient $\binom{M_0}{M_0/2}$ divides this least common multiple: for each prime its valuation is a sum of terms $\lfloor M_0/p^k \rfloor - 2\lfloor (M_0/2)/p^k \rfloor \in \{0, 1\}$. But that coefficient is at least $2^{M_0}/(M_0 + 1) \geq 2^{M_0/2}$, whereas $M_0/2 > r \log_2 M_0$. This is a contradiction. Some prime at most M_0 therefore avoids D , and trial generation and divisibility testing are polynomial. For $r = 0$, $D = 1$ and modulus two suffices. A bound merely on the bit length of the final modulus would not establish this search bound.

The certificate stores $O(sd(r + 1) + t \log(t + 1))$ bits, including the dense decoder and indices. Sparse fraction-free elimination is performed separately on algebraic column blocks. With $m_A \leq 6t$ original matching rows, a conservative dense estimate is $O(m_A sr)$ arithmetic operations on polynomial-size minors. Schoolbook integer arithmetic gives a polynomial bound, for example $O(t^5)$ for the elimination part; source input and output decoding are additional. This coarse upper bound is deliberately not a prediction of measured time.

For a recovered component, the checker computes Nw , checks every exact division by D , and verifies $0 \leq y_i \leq x_i$. The numerators have $O(B + r + \log(d + 1))$ bits. Arbitrary integer weights need not decode integrally: the divisibility test is essential. Reusing a valid decoder on another source with the same positive support preserves correctness, but its original minimum-cost claim need not survive changed coordinate magnitudes.

Corollary 3.4 (Near-optimal reuse under uniform scaling). *If a d -coordinate projection is minimum-slot for x , its slot cost for mx , with integer $m \geq 1$, exceeds the new optimum by at most d bits.*

Proof. For every positive coordinate, $\text{bits}(mx_i) = \text{bits}(x_i) + \text{bits}(m) - 1 + \varepsilon_i$ with $\varepsilon_i \in \{0, 1\}$. Compare the old optimal set J with a new optimal set J' . Both have d coordinates, so the common term $d(\text{bits}(m) - 1)$ cancels. Old optimality gives $\sum_J \text{bits}(x_i) - \sum_{J'} \text{bits}(x_i) \leq 0$, and the difference of the two epsilon sums is at most d . The claimed bound follows. $\square$

4 One scalar weight, with an invariant throughout transport

Order $J = (j_1, \dots, j_d)$ and set

$$\sigma_1 = 0, \quad \sigma_k = \sum_{\ell < k} b_{j_\ell}, \quad L = \sum_{k=1}^d b_{j_k}, \quad E(y) = \sum_{k=1}^d y_{j_k} 2^{\sigma_k}. \quad (3)$$

For every integer $0 \leq y \leq x$, the digits of $E(y)$ recover y_J without carries. Furthermore $K = E(x) < 2^L$. Assign weight 2^{σ_k} to one designated corner of each normal disc of type j_k , and zero otherwise. The weight of a component is exactly $E(y)$.

This construction is an additive linear weight before it is a digit encoding. That distinction permits use of the maintained weighted interval algorithm. It also lets us prove more than final-output correctness.

Theorem 4.1 (Packing preserves live weight runs). *Run vector-valued and packed scalar-valued weighted transport with the same retained coordinates and the same interval-orbit trace. At every stage each live representative carries the sum of original weights over a fibre of original points. Distinct live fibres and emitted completed orbits are disjoint. Every live retained vector is consequently in $[0, x_J]$, its scalar is in $[0, K]$, and the two representations have exactly the same canonical constant-weight run partition.*

Proof. Initially each fibre is a singleton. Changes to the pairing description preserve fibres. When a truncation or fold removes points, the weight of a removed fibre is transferred to its retained image. The operation unions disjoint fibres; translation and reflection formulas calculate these unions without enumerating the points. Completed static orbits are emitted and removed. Induction gives a partition of the original contributions into live and emitted fibres.

There are exactly x_{j_k} original unit contributions of coordinate j_k . No fibre can contain more than all of them. Hence each live vector lies in the box where binary packing is injective. Additivity gives the scalar as E of that vector and conserves total mass K . Two adjacent live vectors are equal if and only if their scalar encodings are equal. Canonical merging therefore creates precisely the same run boundaries in both representations. $\square$

The code's signed difference arrays are an implementation device; they are not live component vectors. A conservative bound on their temporary values is a polynomial in the number of runs and endpoint counts times K . With R runs and initial interval size N_0 , their bit length is $O(L + \log(R + 1) + \log(N_0 + 1))$. Compressed weighted-orbit analysis remains polynomial in the encoded input parameters [2].

This theorem isolates two effects. Switching from the old $q + a$ observer to the support observer can change run counts. Switching from the new d -vector observer to its packed scalar cannot change them. Any runtime difference in the second comparison comes from arithmetic representation and allocation, not a different combinatorial orbit problem.

Packing does not prove a factor- d bit-operation improvement: an addition on an L -bit integer can cost as much as the relevant collection of shorter additions. It can nevertheless reduce Python tuple creation, per-coordinate loops, and hashing overhead. Basis construction, exact decoding, and proof replay must be included when assessing a complete query. The old sparse decoder can be preferable when d is not much smaller than $q + a$.

5 A sharper bound on the number of component types

For a quotient vertex v , let ℓ_v be its connected vertex-link vector. Let a count the vertices for which $\text{supp}(\ell_v) \subseteq S$. These are exactly the vertex-link types occurring in $F(x)$: their common minimum triangle height can be realized by small disjoint links, while a missing required triangle forbids the type. They are two-sided. Every connected triangle-only normal surface is a vertex link under the vertex-link hypotheses.

Lemma 5.1 (Independence of disjoint two-sided types). *Distinct full normal-coordinate vectors of mutually disjoint connected two-sided normal surfaces are linearly independent over $\mathbb{Q}$.*

Proof. An alleged rational dependence can be cleared to an integer one. Nonzero normal vectors have nonnegative entries, so a nontrivial dependence has coefficients of both signs. Rearranging gives equality of two positive integer combinations of distinct vectors. Because the surfaces are two-sided and mutually disjoint, each combination is realized by a disjoint union of the indicated numbers of parallel copies. Equal admissible normal vectors determine normally isotopic surfaces, and that isotopy gives equal multisets of connected full-coordinate types. The two disjoint collections of types in the dependence cannot have equal multisets. This is a contradiction. $\square$

Theorem 5.2 (Support-nullity profile bound). *For every supplied normal vector in the stated manifold setting,*

$$H(x) \leq 2d - a. \quad (4)$$

If all its components are two-sided, then $H(x) \leq d$.

Proof. The two-sided assertion follows immediately from lemma 5.1, since every component vector belongs to the d -dimensional supported space.

For the general assertion retain the a link types separately. For each nonlink type choose one actual component of $F(x)$. A two-sided representative stays as it is. Replace a one-sided representative by the boundary of a small normal interval neighbourhood, with vertical boundary in the ambient boundary. Its connected two-sided horizontal frontier has vector twice the original vector. Neighbourhoods can be chosen disjoint from all other selected representatives and from the vertex links. Coalesce equal resulting vectors, and let h be their number.

Each resulting nonlink vector z has at most two original preimages: the two-sided type z and the one-sided type $z/2$. It cannot become a link type, because its positive quadrilateral coordinates survive doubling. Thus $H(x) - a \leq 2h$. The h distinct resulting types together with the a links are independent by lemma 5.1, so $h + a \leq d$. Combining gives $H(x) \leq a + 2(d - a) = 2d - a$. The empty source satisfies the statement by convention. $\square$

This refines the proof mechanism already used in the repository's `synthesis/compact_coordinates.tex`; it replaces a quadrilateral-count dimension estimate by the exact supported nullity. The factor two is needed. In the one-tetrahedron solid-torus fixture, the primitive one-sided Möbius band has vector $M = (0, 0, 0, 0; 0, 1, 0)$. Its source $3M$ has the central Möbius band and its connected double as two coordinate types, with $d = 1$ and $a = 0$. Adding the disjoint vertex link $\ell = (1, 1, 1, 1; 0, 0, 0)$ gives $(d, a, H) = (2, 1, 3)$. These specific configurations are checked in the supplied finite fixtures; primitivity of an arbitrary Möbius band does not imply support nullity one.

The bound concerns unmarked coordinate inventories. It does not bound the number of ways marked boundary arcs, parallelity data, or future hierarchy attachments can distinguish otherwise identical components. Using it as a bound on global branch count requires a further theorem about those markings.

6 Geometric blocks and exact essential-disc counts

6.1 A conservative partition from original face incidences

Build a graph whose vertices are positive normal-disc types. For every original face-arc equation, connect all its positive participating types before like terms are collected or cancelled. Let $S_1, \dots, S_b$ be the connected blocks, and let $x^{(j)}$ be x restricted to S_j and zero elsewhere.

Theorem 6.1 (Geometric support blocks). *Each $x^{(j)}$ is an admissible matching vector, and*

$$F(x) = \bigsqcup_{j=1}^b F(x^{(j)})$$

up to normal isotopy. The corresponding supported matching spaces form a direct sum, so $d = \sum_j d_j$.

Proof. All positive occurrences of a single original matching equation belong to one graph block. Restricting to any block therefore either keeps that whole equation on the source or sets both sides to zero. Matching and quadrilateral constraints follow.

An actual connection of normal discs across a face pairs arcs appearing in one original face equation. Their types lie in the same block. Following a connected component never leaves its block. Within a tetrahedron the normal discs themselves are disjoint, and the block restriction keeps exactly the selected subcollection. This proves the asserted disjoint union. Finally, no collected matching equation has supported variables in different blocks, which proves the direct sum for the linear spaces. $\square$

Algebraic column blocks are sufficient for matrix elimination. The geometric proof here deliberately uses the more conservative original occurrences. For example, $u + v = u + w$ simplifies to $v = w$, but interpreting how the cancelled type u attaches is a geometric question about the original arc intervals. This algebraic pattern alone is not a counterexample to a finer decomposition. The producer uses a union–find traversal of raw occurrences; the independent checker uses its own graph traversal. Their soundness follows directly from theorem 6.1, without a separate geometric cancellation argument.

6.2 The rank-one shortcut, with the one-sided case included

Lemma 6.2 (Primitive ray is connected). *If $x \neq 0$ and $\dim \mathcal{K}_S = 1$, write $x = mv$, where $m = \gcd\{x_i : i \in S\}$ and v is primitive. Then $F(v)$ is connected.*

Proof. Every component vector of v belongs to $\mathbb{Q}v$ and is a positive rational multiple of v . Primitivity supplies integers c_i with $\sum c_i v_i = 1$. If λv is integral, then $\lambda = \sum c_i (\lambda v_i)$ is integral. The component multiples are therefore positive integers summing to one. There is exactly one component. $\square$

This elementary proof is included to make the specialization self-contained; the vertex-surface principle is classical [1]. It does not say that $F(mv)$ has m components for all primitive rays. If $F(v)$ is one-sided, $F(2v)$ is its connected two-sided frontier. The disc count has a more useful homogeneity property than the total component count.

Let $D(x)$ denote the number of disc components of $F(x)$ whose boundary is essential in the boundary torus. The maintained boundary basis gives two mod-two intersection coordinates for the total boundary class.

Theorem 6.3 (Complete disc count on a supported ray). *Under the hypotheses of lemma 6.2,*

$$D(x) = \begin{cases} m, & \chi(v) = 1 \text{ and } [\partial v] \neq 0 \text{ in } H_1(\partial \mathcal{T}; \mathbb{F}_2), \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Proof. If y is a disc component of x , then $y = kv$ for a positive integer k by the same primitivity argument. Euler characteristic gives $1 = k\chi(v)$, so $k = 1$ and $\chi(v) = 1$. Essential boundary gives the nonzero mod-two class: an essential simple closed curve on a torus has primitive slope and therefore cannot have both slope coordinates even.

Conversely, $F(v)$ is connected. A compact connected surface with Euler characteristic one and nonempty boundary is a disc; the closed alternative with Euler characteristic one is a projective plane, excluded here by the nonzero boundary class. The boundary of this disc is essential because its class is nonzero. A disc is two-sided in the orientable ambient manifold, so $F(mv)$ is m parallel copies. These observations prove both cases. $\square$

Apply the theorem independently to every geometric block with $d_j = 1$. For blocks of larger nullity call the maintained complete disc-count kernel. The total is exact by theorem 6.1. A shared orbit-cycle allowance is charged only to these fallback calls. If it is exhausted, the result is INCONCLUSIVE; a partial count is never a negative recognition verdict. With allowance zero, any number of algebraically certified ray blocks can still be handled.

7 Unit-pivot peeling: an integral ray certificate

The general decoder can spend more work certifying a simple ray than is needed to classify it. Normal matching equations have at most four entries, all units after collection. Their sparse implication structure provides a stronger sufficient certificate.

Definition 7.1 (Unit-peeling transcript). A transcript consists of a seed $j \in S$ and an ordering $i_1, \dots, i_{s-1}$ of $S \setminus \{j\}$. For each h it records an original matching row whose coefficient at i_h is $+1$ or -1 and whose other nonzero supported coordinates are all in $\{j, i_1, \dots, i_{h-1}\}$.

Theorem 7.2 (Integral unit-exposed ray). *Let A be an integer matrix and let $x > 0$ be an integral vector in $\ker A$. If a complete unit-peeling transcript exists from seed j , then*

$$\ker_{\mathbb{Q}} A = \mathbb{Q}p, \quad \ker_{\mathbb{Z}} A = \mathbb{Z}p, \quad p = x/x_j, \quad p_j = 1, \quad \gcd(x_i) = x_j. \quad (6)$$

The primitive generator can be reconstructed by setting $p_j = 1$ and evaluating the recorded rows with integer linear combinations. For normal matching rows these require only additions and sign changes.

Proof. Restrict the recorded rows to the nonseed columns in transcript order. The result is triangular with diagonal entries ± 1 . Thus $\text{rank } A \geq s - 1$. The nonzero vector x gives $\text{rank } A \leq s - 1$, proving equality.

At each recorded row the new coordinate is an integer linear combination of already determined coordinates, because its pivot is a unit. Inductively every coordinate of a kernel vector is an integer multiple of the seed when that seed is integral. Applying the recurrence with seed one gives a vector p and applying it to x gives $x = x_j p$. Hence p is positive and integral; it is in the full kernel because it is x/x_j . Its seed is one, making it primitive. Any rational kernel vector is a rational multiple of p , and its integer seed is exactly that multiple if the vector is integral. This proves the integer-kernel and gcd statements as well as the reconstruction procedure. $\square$

This certificate proves more than rank one. The selected nonseed minor is unimodular, so no denominator, generic modular rank replay, or Euclidean gcd scan is needed on this branch. Both producer and checker enforce unit pivots explicitly; an equation such as $3u = 2v$ cannot be used as a unit-pivot step.

7.1 Discovery, replay, and cost

The producer chooses a seed minimizing $(\text{bits}(x_j), j)$. For every matching row it maintains the count and exclusive-or of as-yet-unprocessed supported variables, together with a per-column incidence list. Process known columns in a queue. Removing an incidence updates the row's

count and exclusive-or; when one unknown remains and its coefficient is a unit, expose it and append the row index and variable index to the transcript. Each column and incidence is processed once. Building the supported restriction in the implementation scans the original coordinate and matching arrays in $O(t)$ structural work; the queue then uses $O(s + \text{nnz}(A))$ work. Their total is $O(t)$ for normal matching matrices, after finite manifold preparation.

The checker does not rerun this discovery algorithm. Starting with the certificate's seed, it inspects the stated original row at each step and checks that the sole unknown is exactly the stated variable with a unit coefficient. It verifies complete support coverage and the source matching equations. Since normal rows have at most four entries, successful evidence contains $s - 1$ indexed steps and occupies $O(t \log(t + 1))$ bits. Source geometry and coordinates remain part of the input, outside this small rank witness.

Primitive recurrence evaluation uses a constant number of additions per step. For a valid positive source the resulting coefficients are at most the source coordinates, so $O(tB)$ bit operations suffice with ordinary addition, after finite geometry has been prepared. Checking source matching also uses $O(tB)$ arithmetic. Including sums of normal-cell contributions gives $O(t(B + \log(t + 1)))$ numerical bit work on one successful block. The implementation retains exact source binding and finite geometry checks. These costs cannot be omitted from a public-API benchmark merely because the rank transcript is short.

If exposure stalls, it establishes no rank lower bound beyond the successful steps. A nullity-one matrix can have no row with a single unknown after one seed is fixed. Our method therefore falls back to the general certified compiler; it does not classify stalled inputs as non-rays. Seed choice here is a deterministic sufficient heuristic, not an optimal stopping-set theorem.

8 An all-family theorem for dense Fibonacci meridians

Consider the repository fixture `layered_torus(t)` for $t \geq 1$. It builds a layered solid torus and its meridian normal vector. Number tetrahedra from 0 to $t - 1$, and write the three positive coordinates in tetrahedron i as $a_i = T_{i,0}$, $b_i = T_{i,1}$, and $c_i = Q_{i,2}$. With $F_1 = F_2 = 1$, the source has

$$a_i = b_i = F_{t-i+1}, \quad c_i = F_{t-i}. \quad (7)$$

All t quadrilateral positions are occupied. The face-pairing permutations in the fixture give the following supported matching equations for $i < t - 1$:

$$b_i = b_{i+1} + c_{i+1}, \quad c_i = a_{i+1}, \quad (8)$$

$$a_i = a_{i+1} + c_{i+1}, \quad c_i = b_{i+1}, \quad (9)$$

and at the terminal tetrahedron $a_{t-1} = b_{t-1} = c_{t-1}$. These are equations from the actual gluing data, rather than an inferred recurrence fitted to timings.

Theorem 8.1 (Linear ray evidence for every layered Fibonacci source). *For every $t \geq 1$ and integer $m \geq 1$, the normal vector $mx^{(t)}$ from eq. (7) admits the implementation's single-seed unit-peeling certificate with exactly $3t - 1$ discovery steps. Its seed coordinate is m , its supported matching nullity is one, and it has exactly m essential-disc components. No interval-orbit trace or Gaussian elimination is required to certify this disc count.*

Proof. For $t = 1$ the minimum-bit/index seed is a_0 , and the terminal equations expose the other two variables. For $t \geq 2$ the first coordinate of minimum bit length is $c_{t-2} = m$. The equations $c_{t-2} = a_{t-1} = b_{t-1}$ expose two terminal variables, and the terminal equation exposes c_{t-1} . Once a terminal triple is known, eq. (9) exposes the previous triple using unit pivots. Induction reaches every positive coordinate. Exactly one seed and $3t - 1$ newly exposed variables occur.

The minimum choice is unchanged by common scaling: the next larger Fibonacci value is at least twice the minimum, so has strictly greater bit length after scaling.

Theorem 7.2 gives nullity one and primitive divisor m . To establish the topology for all t , use the actual normal-cell counts. At height $h = t - i$, the weights on local edges $(01, 02, 03, 12, 13, 23)$ are

$$(F_{h+3}, F_{h+2}, F_{h+1}, F_{h+1}, F_{h+2}, F_h).$$

At $t = 1$ the three global edge weights are F_2, F_3, F_4 . Adding an outer layer keeps the old edge classes and adds the next Fibonacci weight, as follows directly from eq. (9) and the paired faces. Thus the global edge weights are $F_2, \dots, F_{t+3}$ and their sum, the number V of normal-surface vertices, is $F_{t+5} - 2$.

The number of normal discs is $F_{t+5} - 5$. Counting three sides for each triangle and four for each quadrilateral gives total local side count $6F_{t+3} + 4F_{t+2} - 16$. The two boundary faces contribute $2F_{t+3}$ unpaired normal arcs. Consequently the number of surface edges is

$$E = \frac{6F_{t+3} + 4F_{t+2} - 16 + 2F_{t+3}}{2} = 2F_{t+5} - 8,$$

and $\chi = V - E + (F_{t+5} - 5) = 1$.

The boundary torus has one vertex and three edges, with weights $F_{t+1}, F_{t+2}, F_{t+3}$. Their parities define the cocycle dual to the normal boundary. Adjacent Fibonacci numbers are coprime, so these parities are not all zero. On this one-vertex triangulation every vertex coboundary is zero, hence the cocycle is nonzero in cohomology. Duality gives $[\partial x^{(t)}] \neq 0$. Now theorem 6.3 gives exactly m essential discs, with no assumption about the fixture's descriptive name. $\square$

The number of elementary pieces in the primitive source is

$$\sum_{i=0}^{t-1} (a_i + b_i + c_i) = F_{t+5} - 5.$$

It is exponential in t , whereas the rank evidence has $3t - 1$ steps. The binary source itself has $\Theta(t^2 + t \log(m + 1))$ bits, so the theorem is not a linear-bit-time claim for reading the entire input. The certificate's indexed structural part has $O(t \log(t + 1))$ bits, in addition to the encoded source and multiplicity. A dense one-dimensional decoder contains $\Theta(t^2)$ coefficient bits on this family; unit exposure avoids that representation as well as the generic elimination producing it.

The experiment records the maintained orbit algorithm's actual cycle and transmission counts for comparison. These observed counts are not promoted to a theorem about that implementation's schedule for all t . The exact $3t - 1$ count above is independently derived for the complete family.

9 Implementation, independent replay, and source correspondence

The patch is additive. It leaves the default recognizer policy unchanged while exposing the new observer, exact ray count, and source-bound positive certificate as explicit APIs. This makes the work reviewable and avoids turning an overhead-sensitive optimization into an unmeasured global default.

9.1 Module boundaries

Addition under <code>fastunknot/</code>	Responsibility
<code>normal_support.py</code> <code>normal_support_verify.py</code>	Cost-ordered exact support compilation; independent decoder identities and modular rank replay.
<code>normal_support_peeling.py</code> <code>normal_support_peeling_verify.py</code>	Queue discovery of a unit-exposed ray; independent original-row transcript checking.
<code>normal_packed_components.py</code> <code>normal_packed_verify.py</code>	Selected corner weights, scalar or vector transport, full-coordinate recovery, and independently reconstructed packing and classification.
<code>normal_ray_blocks.py</code> <code>normal_ray_blocks_verify.py</code>	Raw-incidence geometric blocks, peeling-first rank gate, exact primitive disc count, and certified fallback.
<code>normal_support_diagram.py</code>	Canonical diagram-exterior binding plus positive disc evidence, with a separate verification entry point.

The support producer uses sparse fraction-free elimination; the checker reconstructs the source matrix and checks the resulting identities. The packed checker builds its own slot radices and corner weights, extracts digits by quotient/remainder arithmetic, evaluates the rational decoder independently, and recomputes the full coordinate summary. It does not call the producer’s weight builder, decoder, or classifier. It additionally checks

$$\sum_{\text{profile entries } (y, \mu)} \mu y = x.$$

This conservation check is useful against omissions and wrong multiplicities; connectivity itself comes from the independently replayed orbit proof.

For geometric blocks the producer uses union-find, while the checker constructs an adjacency graph and traverses it. On a peeled block the producer evaluates unit rows into an array; the checker uses an independently written map of known coordinates. Both validate the recovered primitive normal vector. They establish $x = x_j p$ by matching-space uniqueness and seed values. No large multiplication or per-coordinate division is needed to re-establish that equality. The generic rank-one branch retains exact gcd and division.

The current block interface constructs a full $7t$ vector and revisits finite coordinate geometry for each of its b blocks. This can contribute $O(bt)$ structural work even when every block is individually easy. The linear peeling statement concerns one prepared block, not an unconditional linear bound for this entire public interface. Sharing one sparse geometry pass across blocks is a concrete further optimization.

9.2 Exact integers and failure semantics

Certificates support the maintained hexadecimal integer codec. A JSON string carrying an exact hexadecimal integer is decoded before use; booleans are not silently treated as integers. Schema keys, index uniqueness, dimensions, support completeness, divisibility, coordinate bounds, and counts are checked. Large multiplicities remain integers with binary-length storage. There is no loop over represented sheets or over repeated equal component records.

A user callback supplies cooperative cancellation. Tests inject specific exception objects at geometry, rank, weight, and source-correspondence boundaries and require the same objects

to propagate. A checker returning false for malformed evidence must not accidentally swallow cancellation because its exception class resembles a geometry or proof-validation error.

The public meaning of each outcome is as follows:

Outcome	Claim made
Complete profile	Exact component vectors and multiplicities of this supplied normal surface.
Complete disc count	Exact number of essential-disc components of this supplied normal surface.
Zero disc count	This source contributes no essential disc; another source may do so.
Cycle exhaustion	The requested calculation is inconclusive.
Source-bound positive	A checked essential disc exists in the canonical exterior of the supplied classical knot diagram.

9.3 Binding a disc to a knot diagram

An abstract torus-boundary manifold containing an essential disc is not by itself evidence about a particular diagram. The pinned recognizer already supplies the missing source layer: its Weeks-based diagram exterior uses a canonical pulling subdivision with $20n$ tetrahedra for an n -crossing diagram when $n \geq 1$ (and an alternative centered subdivision), together with an independent local-cell checker. The zero-crossing circle is represented by a Reidemeister-I curl, so the general size formula is $20 \max(1, n)$ for n crossings. The new wrapper uses this maintained correspondence, documented in `synthesis/diagram_exterior.tex` and its linked primary Weeks construction [6].

The function `certify_diagram_ray_disks` constructs that exterior, checks the supplied coordinate vector, obtains and checks a ray-block disc certificate, and returns UNKNOT only if the count is positive. Its certificate contains the source PD code, canonical triangulation, source coordinates, and disc evidence. The verifier checks the PD binding and the exterior independently before replaying the disc proof. For a classical knot in S^3 , compressibility of its exterior boundary implies that it is the unknot. The canonical source construction is part of this implication's input contract.

This interface does not implement disc discovery. Its stabilization examples use coordinates supplied by the existing cocycle seed routine. The ordinary recognizer can already settle many such examples through simpler routes, so timing the wrapper is evidence about a certificate stage, not a speedup of the entire recognizer on those diagrams.

9.4 Trust and resource boundaries

The independent checkers share the finite triangulation representation, normal-corner geometry, exact integer codec, and the established interval proof contract. They do not share producer elimination, greedy discovery, or the ray block planner. The trust boundary therefore remains explicit: the work checks the new optimized representations against maintained exact mathematics; it does not rebuild every existing foundation twice.

The implemented validators are mathematical proof checkers, not resource sandboxes for arbitrary hostile byte streams. Their polynomial bounds are in the lengths of the source and the supplied evidence, subject to the stated finite geometry contract. A future network-facing service would separately need parser and resource limits. Such deployment is outside this additive research package.

10 Correctness audits and performance experiments

The evidence supports a substantial improvement in the ray disc-count query. It also shows that minimizing transported coordinate bits does not guarantee a faster complete census. This section reports both outcomes. All numbers are generated from the retained JSON records; the scripts, raw samples, certificates, input vectors, and SHA-256 source manifests accompany the article.

10.1 Correctness against an independent geometric implementation

The corpus contains 48 validated triangulations and 1,801 supplied standard vertex vectors drawn from the pinned incoming support-kernel corpus. We augment these with zero vectors and deterministic compatible positive combinations, removing duplicates within each triangulation. The resulting 2,980 sources include layered solid tori, boundary caps, interior vertices, Pachner-modified examples, trefoil and figure-eight exteriors, and solid-torus connected sums with $\mathbb{RP}^3$ and $S^2 \times S^1$.

For every source the audit compares the maintained full-coordinate census, the new vector observer, and its packed realization. It independently replays both new profile certificates and the ray-block disc certificate. It also constructs the actual connected components in Regina 7.4, compares their full vectors and multiplicities, and uses Regina’s compressing-disc test on those connected components. Literal component construction is restricted to these small audit inputs; the huge scaling tests never expand normal sheets.

Audit item	Result
Triangulations	48
Supplied normal sources	2,980
New/baseline result comparisons	8,940 agreements
Independent certificate replays	8,940 accepted
Regina full-profile and disc-count comparisons	2,980 agreements
Deliberately mutated profile proofs	27 rejected
Focused new and affected regression tests	146 passed
Additional exact rational matrix audit	40 systems, 420 bases

The nullity distribution is 48, 2091, 524, 217, 90, 8, 2 sources for $d = 0, 1, 2, 3, 4, 5, 6$, respectively. Among 2,932 nonempty sources, 2,117 are settled entirely by ray blocks. Across all sources, 1,644 blocks use peeling, 499 use the generic rank-one certificate, and 815 invoke the complete fallback. These are coverage counts for this constructed, vertex-vector-heavy corpus, not estimates for arbitrary knot diagrams. Both profile inequalities and the identical vector/packed run statistics hold on every audited source.

The 146-test focused suite contains 42 new test methods and 104 affected maintained tests. All pass without skips in the recorded Regina-enabled run. The suite covers the code changed by this additive package and its mathematical dependencies; it is not represented as a rerun of every historical research test in the repository. A separate dense rational oracle checks the greedy minimum against exhaustive eligible bases on 40 small integer systems, independently of the producer’s fraction-free and checker’s modular methods. The unit tests add 100 randomized exhaustive comparisons.

Adversarial checks alter ranks, minor rows, kernel entries, denominators, supports, scalar weights, component multiplicities, primitive divisors, boundary parities, sources, and schema fields. They exercise valid composite moduli with unit pivots and reject invalid nonunit pivots. Several tests disable producer functions during verification. One-sided Möbius bands, closed projective planes, source vectors with more than 20,000-bit common multiplicity, exact

Table 2: Fibonacci meridian disc counts. Both methods include fresh generation and one external replay; the ray producer also performs its existing internal replay. The 512-tetrahedron entry is a ray-only capacity measurement, with no baseline ratio.

t	Pairs/calls	Old disc (ms)	Ray disc (ms)	Old/ray
8	5	3.603	2.428	1.484
16	5	8.400	4.597	1.830
32	5	23.453	8.409	2.733
64	5	82.226	19.655	4.279
128	5	282.173	36.260	7.524
256	3	1351.121	72.965	18.704
512	3	—	227.015	—

hexadecimal transport, and callback identity propagation target failure modes that ordinary meridian examples would miss.

10.2 Timing protocol and reproducibility limits

The final run uses CPython 3.12.14 on Linux x86-64, with garbage collection enabled. Each arm has one retained warm-up followed by five paired rounds; larger cohorts use three. An initially seeded shuffled order is reversed in the next round. A/A arms run the same implementation twice under distinct labels to expose timing variation. Inputs, JSON serialization, result comparison, and artifact writing are outside the timed region.

Every timed public query includes fresh source validation, certificate generation, and one separately timed independent external verification. The new ray producer also performs its own internal replay, which remains inside generation time. Thus its reported time includes *more* replay than the old disc producer; no cost is subtracted to equalize this implementation choice. The source-bound wrapper includes two internal disc replays and a further timed external source-bound replay. A sampled cancellation/time-limit callback is active in every arm; its cost is included.

The run completes 27 cohorts and 652 calls: 538 measured and 114 warm-up calls. There are no timeouts, errors, or budget skips, and all source hashes match before and after. The protocol predeclares the omission of old-query comparisons at 512 tetrahedra; that size is only a new-ray capacity and A/A case. The timeout limits are retained in the script for reproducibility, not used to censor any measured pair.

Each reported speedup is the median of the within-round old/new ratios. The time columns are medians computed separately for each arm, so their quotient need not equal the paired median ratio. Ratios above one favour the new method. Across all A/A controls the median-ratio range is 0.862928–1.093790. Small differences within this range should not be treated as robust improvements, and three large-case rounds do not supply a general statistical confidence interval. Measurements are single-host observations with recorded instrumentation, not universal hardware constants.

Table entries are separate medians in milliseconds. Each ratio is the median of the first time divided by the second time within the same measured round; it need not equal the ratio of the two displayed medians. Warm-ups are excluded.

10.3 The successful specialization: disc counts on Fibonacci rays

Table 2 reports the principal gain. At 256 tetrahedra, the old disc query has median total time 1351.121 ms and the peeled query has median 72.965 ms. The paired median improvement is 18.704 times, with paired range 17.083–20.191 in the three rounds. The new method establishes

Table 3: Complete coordinate censuses on the same Fibonacci sources: dimension reduction does not compensate for general rank-compilation overhead in this experiment. The old implementation already uses its single-orbit shortcut. There are five pairs through $t = 128$ and three at $t = 256$.

t	$q + a$	d	Old (ms)	d -vector (ms)	Packed (ms)	Old/packed
8	8	1	2.938	3.130	3.143	0.916
16	16	1	7.163	8.576	8.176	0.868
32	32	1	22.625	26.072	26.098	0.867
64	64	1	79.959	97.105	93.013	0.855
128	128	1	288.641	368.977	347.524	0.820
256	256	1	1335.527	1445.574	1707.417	0.782

Table 4: Control cases, five pairs per row. The first and second methods are named in each group heading. Fixed-trace transport uses an identical selected basis and independently checked orbit trace in both arms; its excluded setup contains geometry, compilation, weight construction, and orbit discovery.

Case	First (ms)	Second (ms)	First/second
<i>Disc counts: old kernel / ray interface</i>			
Vertex links	0.398	1.515	0.259
Three-way mixture	1.394	3.323	0.416
Capped meridian	5.568	15.471	0.346
<i>Coordinate censuses: old compact / packed</i>			
Vertex links	2.381	2.551	0.920
Three-way mixture	2.873	3.356	0.917
Capped meridian	18.637	16.711	1.087
<i>Fixed-trace transport: vector / packed</i>			
Vertex links	0.886	0.914	0.953
Three-way mixture	1.003	1.059	1.004
Capped meridian	9.502	9.199	1.042

Table 5: Source-bound certificates for stabilized-circle diagrams. The existing cocycle discovers the supplied vector outside the timers. Generation includes two internal disc replays, followed by a separately timed external source-bound replay. These are new-wrapper A/A stages, not a comparison of complete automatic recognizers. Separate stage medians need not sum to the total median.

Crossings	t	Calls	Generate (ms)	Verify (ms)	Total (ms)	Proof bytes
1	20	5	8.072	2.878	11.023	4,283
4	80	5	24.331	8.686	34.129	16,333
8	160	5	54.233	17.607	70.303	32,823
16	320	3	111.712	45.605	159.955	66,815
32	640	3	255.049	81.645	342.695	134,943

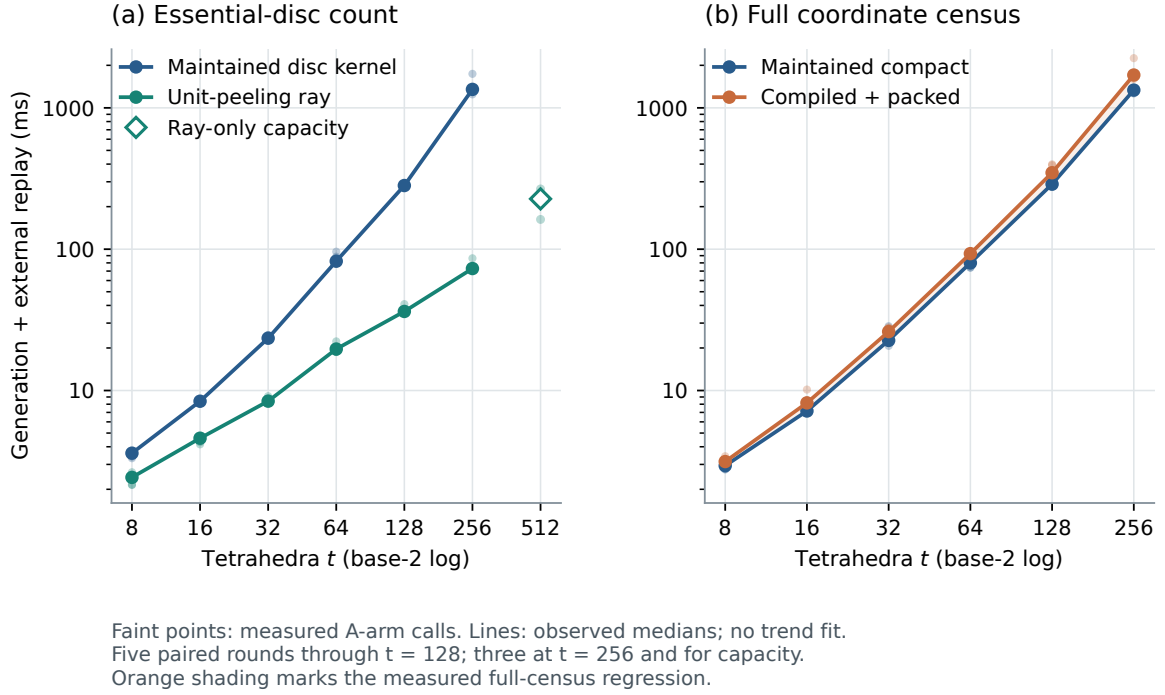


Figure 1: Recorded generation plus independent external verification times. The ray-disc specialization improves strongly on this family; the general packed full-coordinate census does not. Points are measured medians, with no fitted asymptotic curve. The new-ray-only 512-tetrahedron point is a capacity measurement.

the single supported ray and its primitive disc data directly; it avoids the orbit discovery that the old primitive-core query still performs.

At that size the serialized disc certificate shrinks from 7,098,789 to 15,368 bytes, about a factor of 462. Both sizes use the same compact JSON serialization and exclude the separately stored source input. The old certificate records 261 orbit cycles and 81,930 transmissions, while the new one records one peeling block and zero orbit cycles. These counters explain the measured difference without turning an observed scheduling pattern into a theorem about all interval algorithms.

At 512 tetrahedra the new query has median total time 227.015 ms and a 32,264-byte certificate. This establishes tested capacity on that input; it has no measured old/new ratio. The all-family asymptotic statement comes from theorem 8.1, whose $3t - 1$ steps are proved for every t and every positive integer multiplier, independently of these timing samples.

10.4 Negative and inconclusive performance findings

Table 3 is an essential qualification. On the same Fibonacci family the general packed census is slower at every measured size. At $t = 256$ the old census takes median 1335.527 ms and the packed census 1707.417 ms, with old/new paired ratio 0.782. The new projection reduces dimension from 256 to one, but the baseline already invokes its single-orbit shortcut for weighted transport. The common orbit discovery remains, and general rank compilation adds cost. A mathematically optimal slot budget has not become a faster end-to-end query here.

The ray API also regresses on the three overhead controls in table 4: interior links, an interior sphere/disc/Möbius mixture, and a capped meridian with links. The old primitive-core routine disposes of these small cases cheaply. The new conservative block construction, rank gate, repeated geometry scans, and internal replay cost more than they save. This is why the

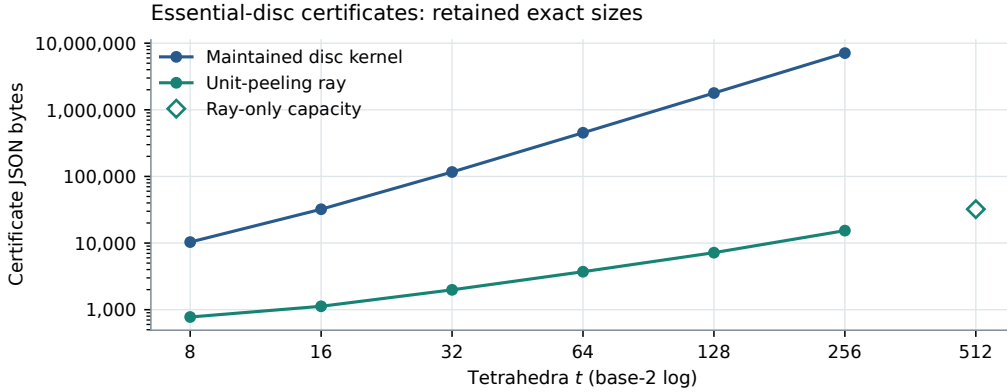


Figure 2: Compact JSON disc-certificate sizes on the same family, excluding source input. The independently replayable unit-row transcript avoids the large retained orbit trace. The 512-tetrahedron diamond has no old-query comparison; connecting lines only link measured values.

additions are explicit APIs, with no automatic replacement of the default policy.

The isolated transport comparison holds the selected basis and unweighted orbit trace fixed. Vector/packed paired ratios are 0.953, 1.004, and 1.042 for the three controls. The last has a broad paired range 0.632–1.701. These observations validate the shared-run comparison but do not establish a robust scalar-packing wall-clock advantage. The capped full census gives a small 1.087 paired improvement, likewise too close to the A/A variation to motivate a broad default change.

The practical conclusion is specific: the peeled rank-one disc gate is a demonstrated improvement on the dense tested family. The general observer provides exact minimal dimension, optimal allocated bits, reusable rank evidence, and a foundation for later work; current measurements do not justify advertising it as a universal acceleration.

10.5 Source-bound diagram stages

Table 5 reports positive braid stabilizations of a circle: the closure of $\sigma_1\sigma_2\cdots\sigma_n$ on $n+1$ strands. The maintained cocycle routine supplies a vector before timing. The measured stage constructs the canonical exterior, generates the new positive proof, and independently checks its source binding. All sizes through 32 crossings and 640 exterior tetrahedra complete. There is an A/A control but no old/new recognizer comparison. These inputs verify the source connection of the new certificate path; they are not evidence that the complete recognizer is faster on hard unknot diagrams.

11 What remains between these results and quasi-polynomial recognition

The new local bounds are useful because they reduce the information transported and sometimes avoid transport entirely. They do not show how many source supports recognition must try. Even if every tested candidate is cheap, exponentially many candidates would remain an exponential recognizer.

The following elementary accounting proposition states the required global interface precisely. It is a conditional composition statement, not a verified property of the present recognizer.

Proposition 11.1 (Conditional hierarchy accounting). *Suppose a complete recognition procedure on an n -crossing input visits at most $L(n)(g(n) + 1)^{L(n)}$ hierarchy stages, where $L(n) \geq 1$ and $g(n) \geq 0$. Charge all preprocessing, terminal work, failed attempts, and proof assembly to these stages. Suppose the total charged work at each visited stage, including candidate selection, compressed updates, intermediate representation, and independent checking, is at most $P(n)$. Then its total time is at most*

$$L(n)(g(n) + 1)^{L(n)}P(n).$$

In particular, if $L(n) = O((\log n)^2)$, $g(n) = n^{O(1)}$, and $P(n) = 2^{O((\log n)^3)}$, the complete procedure takes $2^{O((\log n)^3)}$ time. The same conclusion holds with polynomial P .

Proof. Multiply the number of visits by the cost of a visit. Its logarithm is at most $\log L + L \log(g + 1) + \log P$, which has the claimed order under the stated hypotheses. Work hidden in candidate selection or in an expanding intermediate object must be included in P for the implication to apply. $\square$

The shape of the visit bound occurs in Lackenby’s hierarchy analysis [5, Proposition 9.1]; verifying the stronger parameter hypotheses for an implemented compressed recognizer is a separate research task. Four specific distinctions prevent an incorrect use of proposition 11.1 here.

1. **A supplied vector versus a useful vector.** Support nullity and peeling are measured after the support has been supplied. They do not identify a meridian support among all admissible choices.
2. **A type inventory versus marked continuations.** The bound $H \leq 2d - a$ counts unmarked coordinate types. A hierarchy can distinguish boundary attachments, parallelity regions, and provenance not present in that inventory.
3. **Small output versus cheap construction.** A short rank witness can still require expensive discovery on an arbitrary support; a compressed cut can have short invariants but expensive attaching maps.
4. **A local budget versus a global budget.** Restarting a successful local procedure at every branch does not control the sum of its work. A monotone potential or a complete stage-count theorem is required.

The diagram-exterior correspondence is already checked for the canonical input construction. The remaining source problem is maintaining comparable provenance through the retriangulations and compressed cuts of a future hierarchy. Treating the input correspondence itself as wholly missing would misdescribe the current repository.

12 Further research questions and proposed theorem targets

The following programme separates near-term algorithmic work from the global theorems that would change the asymptotic recognition result. Each item names an observable object or a falsifiable target. Successful experiments may motivate a theorem but do not replace one.

12.1 Priority A: extend the provable local savings

1. Multi-seed unit elimination with a controlled residual. When a one-seed closure stalls, introduce a new seed and express later exposed coordinates as integral forms in the seeds. The unsolved original rows then give a residual system. Can a seed-selection rule bound the number and width of these forms by a structural parameter that stays small under relevant

layering and cutting operations? A plausible target is a certified reduction to k residual variables in polynomial time with controlled coefficient bits. One must not equate the number of greedy seeds with d : sparse exposure closure is weaker than linear span closure.

2. Characterize and minimize single-seed stopping sets. View the supported matching rows as a bounded-arity incidence hypergraph. Which connected normal supports admit a complete unit-peeling order? Can obstructions be characterized by a small core, and can that core be computed without large integer elimination? Seek families with $d = 1$ but growing unavoidable core size to delimit the shortcut’s applicability. Reordering rows alone cannot solve every stopping-set obstruction.

3. Optimize a measured cost, beyond the slot sum. The present matroid theorem optimizes L , but exact decoder density and coefficient height also cost time. Can one choose an observer with a rigorous bicriteria bound on slot budget and decoder sparsity? A target could compare to the best basis under $L + \lambda \text{nnz}(N)$, or prove an approximation guarantee under a sparse normal-matching structure. The simple greedy proof does not apply automatically to this nonadditive objective.

4. Make general rank compilation reusable under support edits. Hierarchies and candidate families often differ by a few zero coordinates. Can a certified rank/decoder update handle a deletion or insertion of k columns at cost polynomial in k and the affected algebraic blocks, rather than recompiling globally? The verifier should check a small update against the old certified matrix while preserving exact source support binding.

5. Prove a practical adaptive observer policy. Develop a preflight estimate comparing the old sparse $q + a$ decoder, the new d observer, and the ray gate. A useful target is an overhead guarantee: the dispatcher takes at most a small certified budget before reverting to the baseline, and total time is bounded by that budget plus the selected complete query. The negative benchmark cases in this package provide a test set for such a policy. Prediction accuracy alone is not an asymptotic guarantee.

12.2 Priority B: retain geometry and markings under compression

6. A marked refinement of $H \leq 2d - a$. Specify exactly which markings a hierarchy needs on a component: boundary arcs, attachment intervals, orientation data, and maps to the original exterior. Does a finite invariant of controlled bit length determine the allowed next cuts? Seek a bound on distinct marked states in terms of d and marking complexity. A counterexample with fixed d and unbounded necessary markings would be equally valuable. Coordinate equality alone is insufficient for this purpose.

7. Source-bound compressed cutting and gluing certificates. Realize existing compressed-cut mathematics as an executable source-bound operation with an independent checker. Lackenby’s Theorem 14.4 already gives polynomial cutting and parallelity-bundle attachment data for orientable normal surfaces in uniform-type handle structures, in handle count and logarithmic extended weight [5]. The new target is a checked representation in this architecture that carries the required source maps through those operations. The canonical input exterior checker is an existing base case; later cuts require their own maps and verified hypotheses.

8. Stable local handle types and support complexity. Identify the precise class of handle structures produced by an implemented hierarchy and prove that its operations preserve a uniform local incidence bound. Then determine how supported nullity, peeling cores, and marking size change under those operations. A theorem about arbitrary triangulations cannot silently supply a bound for an expanding hierarchy representation.

9. Product-region removal with a charging argument. Can raw-support blocks, ray certificates, and two-sided profile types identify parallelity/product regions that can be removed while charging every removal to a decreasing potential? The target is a global bound on cumulative representation size and total local work, not just termination of each individual removal.

12.3 Priority C: close the global selection and hierarchy bounds

10. Find a useful support without exhaustive support enumeration. Determine whether a meridian or hierarchy surface can always be found within a quasi-polynomially bounded family of supports, after allowed checked geometric preprocessing. Possible parameters include a bounded exposure core, support nullity, and normal boundary data. The dense Fibonacci family shows that few occupied quadrilateral positions are not necessary for easy certification. A low-nullity promise alone is not a discovery algorithm.

11. Quantify the limits of coherent cocycle candidates. The maintained minimum-span and Euler-optimal coherent cocycle family is not a complete meridian family on arbitrary solid-torus triangulations, as proved in `synthesis/coherent_obstruction.tex`. That obstruction does not establish failure on canonical diagram exteriors. Formulate a controlled enlargement that repairs identified obstructions and prove its candidate count and bit-height bounds. Re-running the same optimizer with a new tie-breaker does not address a structural incompleteness theorem.

12. Prove a polylogarithmic hierarchy potential. Seek a nonnegative potential on the complete marked state whose decrease bounds the relevant hierarchy length by $O((\log n)^2)$, or by another polylogarithmic function sufficient for proposition 11.1. It must charge all reset and branching operations. A linear bound in a complexity parameter that may itself grow polynomially in n does not yield this conclusion.

13. A complete bit-complexity ledger. For every stage, bound the number of candidates, dimensions, coordinate heights, decoder/certificate sizes, boundary maps, and repeated local calls. Then compose the bounds without reinitializing the accounting at each branch. The deliverable's source hashes and separated generation/replay timings are small-scale infrastructure for this ledger; the missing proof is global. For negative verdicts, seek shared certificates that exclude whole families of failed support branches after unit substitutions. A linear dual witness can settle an individual convex cone, but a disjunction of quadrilateral sectors requires evidence covering every alternative. A certificate DAG with shared substitution prefixes is a concrete target; a single failed candidate is not a complete rejection argument.

12.4 Priority D: falsification and validation

14. Adversarial geometric families for overhead and core growth. Build genuine torus-boundary triangulations with large d , many supported links, persistent exposure cores, and one-sided components. Compare exact costs before choosing a default policy. Abstract

sparse matrices test algebra but cannot substitute for realizable normal-matching geometries when the claim is geometric.

15. Formalize the smallest trusted lemmas first. The triangular unit-pivot lattice theorem and integer decoder identities are compact candidates for a proof assistant or a very small extracted checker. Next formalize raw-incidence block separation and the primitive ray disc criterion. Keep a traceability map from each formal statement to the concrete certificate fields and finite manifold hypotheses. This would strengthen the independent Python checks without claiming that the present package is already formally verified.

The immediate implementation order suggested by the evidence is an adaptive local policy, multi-seed residual compilation, and a source-bound compressed cut interface. The highest-value mathematical target remains a selection or hierarchy theorem controlling the complete marked state; local arithmetic savings alone cannot supply it.

A Reproduction and integration guide

The archive is organized so that the mathematical report can be read independently and the code can be reviewed as an additive change. The `repo_overlay/` tree contains new repository-relative files; the `integration.patch` file describes the same additions. A compact `reference_snapshot/fast/` contains the pinned maintained Python source, fixtures, and tests needed by these experiments, excluding historical result archives, compiled caches, and unrelated large artifacts. It is a source subset, not a complete Git checkout.

The root reproduction script assembles a fresh run directory from the reference snapshot and overlay, verifies its source provenance, and runs the focused test suite or experiments. It does not modify an existing checkout. For a real integration, first check and apply the patch at the pinned commit, review the nine added modules and five new test files, and run the included commands. The patch changes no existing production file. Updating the repository’s synthesis index or incoming-work inventory is intentionally left to the integration step requested by the user.

Python’s standard library suffices for the new production code, focused tests without optional geometric oracles, the algebra audit, and the benchmarks. Regina 7.4 was used for the recorded geometric oracle run; install it separately to repeat that oracle. It is not bundled. Matplotlib is only needed to regenerate the plots. A normal TeX installation with the listed packages builds the article using two `pdflatex` passes; there is no external bibliography download or shell-escape requirement.

```
# From the unpacked archive root:
python3 reproduce.py --task tests
python3 reproduce.py --task audit --regina
python3 reproduce.py --task benchmark

# Or from an integrated fast/ directory:
python3 support_research/run_tests.py \
  --output support_research/results/local_regression.json
python3 support_research/audit.py --regina \
  --output support_research/results/local_audit.json
python3 support_research/benchmark.py --diagrams 1,4,8,16,32 \
  --output support_research/results/local_benchmark.json
```

The optional `--regina-path` argument points the audit and focused runner at a separately installed Regina module; the runner propagates it to the maintained subprocess integration

tests. Timings vary with hardware and load. Compare exact result hashes, successful replays, recorded scope, and within-run paired ratios before comparing absolute milliseconds.

B Certificate schemas and executable contracts

The following descriptions are field-level guides; the independently written checkers are the executable schema definitions. Integers may use the exact hexadecimal transport supported by the maintained codec.

B.1 Supported matching kernel

```
schema: "normal-support-kernel-v1"
support, selected, pivots, pivot_rows
rank, nullity, denominator, numerators, modulus
```

`support` lists exactly the positive source indices. The `selected` and `pivot` lists partition it, with sizes d and r . The row list identifies the claimed nonsingular original-row minor. The numerator matrix is indexed by support rows and selected-coordinate columns. The checker verifies strict indices, the identity block, all kernel equations, and modular unit-pivot rank. Optimality is a producer theorem; a valid nonoptimal decoder is still valid evidence of the supplied profile calculation.

B.2 Unit-exposed ray

```
schema: "normal-support-ray-peeling-v1"
support, seed, steps, nullity: 1
# Each step: [original_matching_row, newly_exposed_coordinate]
```

The checker reconstructs the source support, checks all source matching equations, validates the seed, requires exactly $s - 1$ steps, and checks the one-unknown unit-pivot condition at every step. Completion establishes the integer ray theorem. No list of primitive coefficients is needed in the certificate; they are regenerated by the row recurrence when topology is queried.

B.3 Packed full-coordinate profile

```
schema: "normal-packed-components-v1"
input_sha256, encoding, support_kernel
boundary_homology_basis, weighted_orbits, summary
```

The encoding is either `packed` or `vector`, using the same selected coordinates. The source fingerprint binds the complete triangulation and coordinates. The orbit subcertificate uses the maintained weighted replay format, whose unweighted trace key is `orbit_proof`. Every decoded component is integral, within source bounds, and classified from its recovered complete vector. The checker validates the total vector sum and all reported multiplicities and disc counts.

B.4 Geometric block disc count

```
schema: "normal-ray-block-disks-v1"
input_sha256, boundary_homology_basis, blocks
compressing_disk_components
```

The block list must equal the independently reconstructed raw-occurrence partition. A ray entry contains its support certificate, primitive divisor, Euler characteristic, two boundary parity values, and disc count. A fallback entry contains the maintained complete disc-count certificate. Counts are checked block by block and summed. No total number of connected components is inferred from the primitive divisor.

B.5 Positive knot-diagram evidence

```
schema: "diagram-ray-disks-v1"
input_pd, triangulation, coordinates, disc_certificate
```

The source checker requires the same normalized PD code, independently verifies its canonical exterior, replays the disc certificate, and accepts only a strictly positive count. The producer returns `INCONCLUSIVE` when the supplied vector has no essential disc. Invalid geometry, invalid coordinates, and corrupted proofs never become a negative knot verdict.

C A compact API example

This example exercises an exponentially growing family while retaining a short input vector and a short rank certificate. It calls both the producer and its independent checker.

```
from normal_orbit_research.fixtures import layered_torus
from fastunknot.normal_ray_blocks import normal_ray_block_disk_count
from fastunknot.normal_ray_blocks_verify import (
    verify_normal_ray_block_disk_certificate,
)

triangulation, primitive = layered_torus(128)
multiplicity = (1 << 5000) + 1
source = [[multiplicity * x for x in row] for row in primitive]
answer = normal_ray_block_disk_count(
    triangulation, source, max_cycles=0, record_certificate=True,
)
assert answer["status"] == "COMPLETE"
assert answer["compressing_disk_components"] == multiplicity
assert answer["stats"]["orbit_cycles"] == 0
assert verify_normal_ray_block_disk_certificate(
    triangulation, source, answer["certificate"],
)
```

This invocation proves a fact about the supplied surface in the supplied solid torus. To assert a fact about a knot diagram, use the source-bound wrapper with coordinates in that diagram's canonical exterior. Supplying coordinates from an unrelated triangulation is not a valid shortcut.

D Proof and evidence traceability

Mathematical claim	Checked object	Main evidence
Minimum coordinate cost	Greedy pivot selection and exact decoder	Exchange proof; exhaustive rational basis audits.
Complete source reconstruction	$AN = 0$, identity rows, nonsingular minor	Independent modular replay and divisibility checks.
Scalar transport equality	Common basis, original contribution fibres	Theorem 4.1; equal run statistics in every corpus case.
Profile bound	Distinct disjoint types and horizontal doubles	Theorem 5.2; sharp one-sided and link fixtures.
Geometric splitting	Original positive face-arc occurrences	Independent adjacency reconstruction.
Primitive connected ray	Rank-one support and integer primitivity	Classical principle with explicit proof; one-sided tests.
Integral unit ray	Triangular original-row transcript	Independent unit-pivot replay; disabled generic paths.
All Fibonacci sizes	Literal gluing recurrence and cell counts	Theorem 8.1; finite implementation tests.
Diagram-positive verdict	Canonical exterior plus positive disc proof	Source mutation tests and canonical exterior checker.
Global quasi-polynomial goal	Full marked search and transition bounds	Conditional accounting only; open research agenda.

The mathematical and software reviews corrected several overly broad draft formulations: a symbolic cancellation pattern is not itself a geometric counterexample; the double of a proper one-sided surface uses a horizontal frontier; small support does not remove the implementation's full-array scan; and coherent-cocycle obstructions on arbitrary solid-torus triangulations do not imply an obstruction on canonical diagram exteriors. These distinctions are retained in the final statements and matter for future integration.

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